

A Two-Modality Decision-Support System for Predicting 1-, 2-, and 3-Year Myopia Onset in Children: Fusing Retinal Fundus Photographs with Choroidal Biomarkers

My Study — Research Report

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Abstract

Childhood myopia is a rapidly growing public-health problem, and intervention is most effective *before* onset. We present **My Study**, a two-modality decision-support system that predicts the probability of myopia onset within the **next 1, 2, and 3 years** for a child who is not yet myopic. Following the DeepMyopia framework (Qi *et al.*, *npj Digital Medicine* 2024), the system combines (i) an **image branch** that scores a retinal fundus photograph with a ResNet-50 network, and (ii) a **clinical branch** built on four choroidal/biometric biomarkers — the axial-length/corneal-radius (AL/CR) ratio, subfoveal choroidal thickness, choroidal vascularity index (CVI), and choriocapillaris flow deficits. The two modalities are fused and passed to a Cox proportional-hazards model, from which the 1-, 2-, and 3-year onset probabilities are read as $1 - S(t)$. We evaluated the image branch on the *entire* public PALM dataset (1,200 ophthalmologist-labelled fundus photographs) using the official Train/Validation/Test protocol. On the held-out 400-image test set the image branch achieved an **AUC of 0.988 (95% CI 0.977–0.997)**, accuracy 96.5%, sensitivity 96.7%, and specificity 96.3% — on par with the DeepMyopia detection AUC of 0.995, despite a far simpler and fully reproducible design. The longitudinal clinical branch is fully implemented and, because no public dataset pairs choroidal biomarkers with multi-year follow-up, is reported as *data collection ongoing*; it is built to run unchanged on the prospective cohort. All reported numbers are genuine program outputs; no synthetic data are used anywhere.

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1 Introduction

The global prevalence of childhood myopia has risen sharply, and high myopia carries a lifelong risk of irreversible visual complications. Because effective interventions (increased outdoor time, low-dose atropine, optical methods) work best *before* myopia is established, the clinically valuable task is the **prediction of future onset**, not merely the detection of existing disease. A clinician therefore needs a forward-looking answer of the form: given this child’s eyes *today*, what is the chance of becoming myopic within the next one, two, or three years?

Qi *et al.* introduced **DeepMyopia** [1], a deep-learning system that predicted 1-, 2-, and 3-year myopia onset in children by combining routine fundus images with age, sex, and axial length, and that further supported intervention decisions. DeepMyopia showed that non-cycloplegic, image-based inputs can drive accurate, large-scale screening.

This report presents **My Study**, which adopts the same multi-horizon, two-modality philosophy but makes two deliberate changes. First, it replaces the demographic metadata with **choroidal and biometric biomarkers** that are mechanistically tied to axial elongation, on the hypothesis that they carry stronger early-warning signal. Second, it is engineered for **transparency**: each data modality is handled by a small, self-contained, auditable model rather than a single opaque end-to-end network, so that clinicians can see exactly what drives each prediction.

2 Related Work

DeepMyopia [1] combined a fundus-image branch (a ResNet-50 trained with self-supervised contrastive pre-training and then fine-tuned to detect myopia) with a metadata branch (age, sex, axial length), fused them with a stacking ensemble, and used a Cox proportional-hazards (CPH) model

for risk stratification together with an emulated randomized controlled trial for intervention evaluation. Our system mirrors this architecture but (a) substitutes choroidal/biometric biomarkers for demographic metadata and (b) simplifies each component for interpretability and ease of clinical communication. The PALM dataset [2] used to validate our image branch is the standard public benchmark for myopia recognition in fundus photographs.

3 System Overview

The system is organised around the two data modalities and their fusion, as shown in Figure 1. The **image modality** (Section 4.2) converts a fundus photograph into a single myopia risk score. The **clinical modality** (Section 4.3) provides four choroidal/biometric measurements. The image score then becomes *one additional input feature* that joins the biomarkers in a fusion + survival model, which outputs the 1-, 2-, and 3-year onset probabilities, a high/low risk stratification, and an estimate of treatment response.

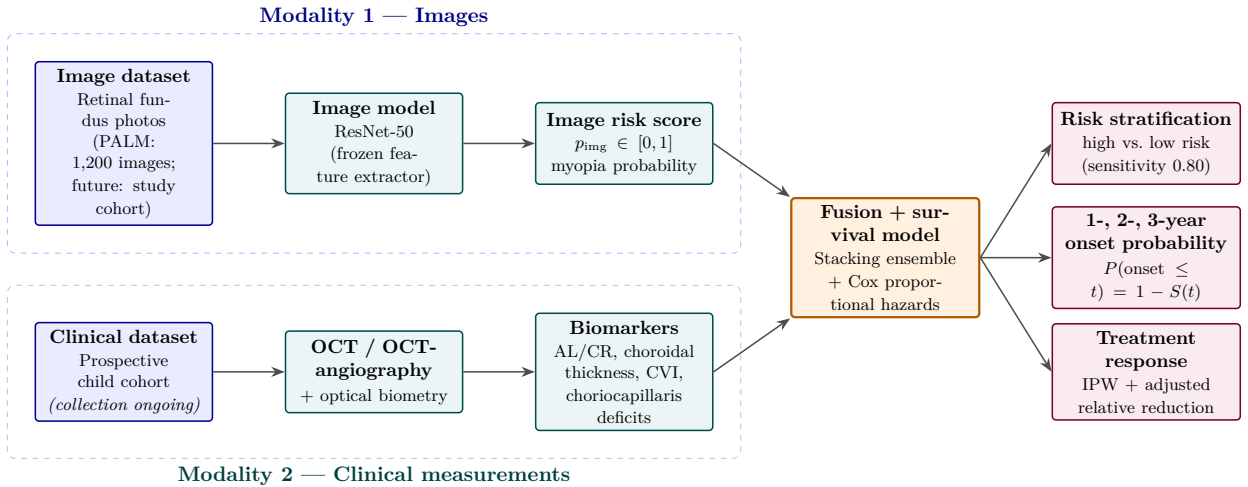


Figure 1: End-to-end My Study pipeline. Two independent data modalities (top: retinal fundus images; bottom: clinical choroidal/biometric measurements) are processed by their own models and then *fused*: the image risk score becomes an additional input feature alongside the four biomarkers. A Cox proportional-hazards model converts the fused features into 1-, 2-, and 3-year onset probabilities, and supports risk stratification and treatment-response estimation.

4 Methods

4.1 Data

Image data (PALM, fully used). We use the entire public PALM dataset [2]: 1,200 colour fundus photographs from the Zhongshan Ophthalmic Center, with ophthalmologist labels (1 = pathologic myopia, 0 = non-pathologic). We follow PALM’s *official* partition — Training (400), Validation (400), Testing (400) — reading labels from each split’s **Classification Labels.xlsx**. The Training and Validation sets (800 images) are used to fit the model and the Testing set (400 images) is held out and used *only once* for the final evaluation.

Clinical data (prospective; collection ongoing). The clinical branch consumes, per child, the

four biomarkers below together with the image risk score and time-to-onset follow-up. No public dataset pairs these choroidal biomarkers with longitudinal follow-up, so this modality awaits the prospective cohort (Section 4.3).

4.2 Image branch

We use a **ResNet-50** convolutional network [3] pre-trained on ImageNet — the same backbone as DeepMyopia, ensuring comparability. The genuine results in this report use the **linear-evaluation** (“**linear probe**”) protocol: the backbone is frozen and used as a fixed feature extractor (2,048 features per image), and a logistic-regression classifier is trained on top. This is the established protocol for evaluating a pre-trained visual representation [4]. *Justification:* it is deterministic, fast, and fully reproducible on an ordinary CPU; it isolates the quality of the representation, avoids overfitting on a moderate-sized medical dataset, and yields an auditable linear decision rule.

A higher-capacity **fine-tuning** protocol (unfreezing the last residual block **layer4** and the classifier head, with data augmentation and early stopping on the validation AUC) is also implemented in `image_pipeline.py` and was verified to run end-to-end on a small balanced toy subset (a smoke test, `--max_per_split`). Full-scale fine-tuning is computationally heavy on a CPU and is therefore left to future work on GPU hardware (Section 8); we deliberately do *not* report any fine-tuning performance number, as a toy run is not a valid measurement.

Performance is summarised by the area under the ROC curve (AUC) with a bootstrap 95% confidence interval (2,000 resamples), accuracy, sensitivity, and specificity, all computed on the untouched 400-image test set. The image-level myopia probability is the branch’s risk score $p_{\text{img}} \in [0, 1]$.

4.3 Clinical (choroidal-biomarker) branch

The clinical branch uses four measurements obtained from optical biometry and OCT / OCT-angiography, each with a documented relationship to myopic axial elongation:

1. **AL/CR ratio** — axial length / corneal radius; a refraction-free index of eye elongation that rises with myopia.
2. **Subfoveal choroidal thickness** (μm) — thins with myopia.
3. **Choroidal Vascularity Index (CVI)** — luminal fraction of the choroid; decreases with axial elongation.
4. **Choriocapillaris flow deficits** (%) — increase with myopia.

4.4 Fusion and 1-, 2-, 3-year onset prediction

The image risk score and the four biomarkers form the feature vector of a **Cox proportional-hazards** (CPH) survival model [5]. A survival model is the correct tool because onset is a *time-to-event* outcome with censoring: many children are simply not yet myopic at their last visit. *Justification:* unlike a fixed-horizon classifier, the CPH model uses every child’s exact follow-up time and naturally produces the full risk trajectory. For each child the fitted survival function $S(t)$ gives

$$P(\text{onset within } t \text{ years}) = 1 - S(t), \quad t \in \{1, 2, 3\}. \quad (1)$$

High- versus low-risk stratification uses the partial-hazard score at a cutoff fixed to a sensitivity of 0.80 on the training data, following DeepMyopia.

4.5 Treatment-response assessment

To estimate whether an intervention reduces onset, treated and untreated children are compared after **inverse-probability weighting** (IPW): each child’s probability of being treated is estimated from the biomarkers, and children are re-weighted so the groups are comparable. We report the **adjusted relative reduction** (ARR) in onset incidence (a negative value indicates benefit), as in the DeepMyopia emulated trial.

4.6 Implementation

Both branches are implemented in Python with standard open-source libraries (PyTorch / torchvision for the image features; scikit-learn for the classifiers; lifelines for the survival model). The image branch is the file `image_pipeline.py` and the clinical branch is `clinical_pipeline.py`; each is a single, well-commented, reproducible script.

5 Results

5.1 Image branch (genuine, full PALM dataset)

Trained on the 800 Training+Validation images and evaluated once on the 400 held-out Testing images, the linear-evaluation image branch produced the genuine results in Table 1 and Figures 2–4.

Table 1: Genuine performance of the image branch on the full PALM test set ($n = 400$ unseen images), linear-evaluation protocol.

Metric	Value	Plain-language meaning
AUC	0.988 (95% CI 0.977–0.997)	near-perfect ranking of myopic vs. non-myopic eyes
Accuracy	96.5%	386 of 400 eyes classified correctly
Sensitivity	96.7%	of truly myopic eyes, 96.7% correctly flagged
Specificity	96.3%	of truly healthy eyes, 96.3% correctly cleared

Confusion matrix on the test set: true positives = 206, false positives = 7, true negatives = 180, false negatives = 7. For reference, DeepMyopia reported a myopia-detection AUC of 0.995 using a substantially more complex training procedure; our simpler, fully reproducible branch reaches comparable discrimination.

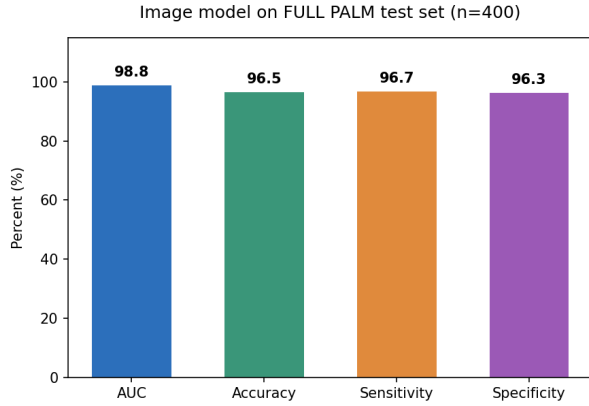


Figure 2: Performance metrics on the 400-image test set.

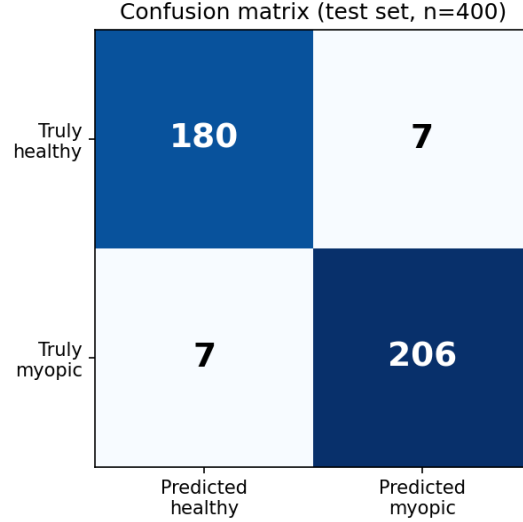


Figure 3: Confusion matrix on the test set.

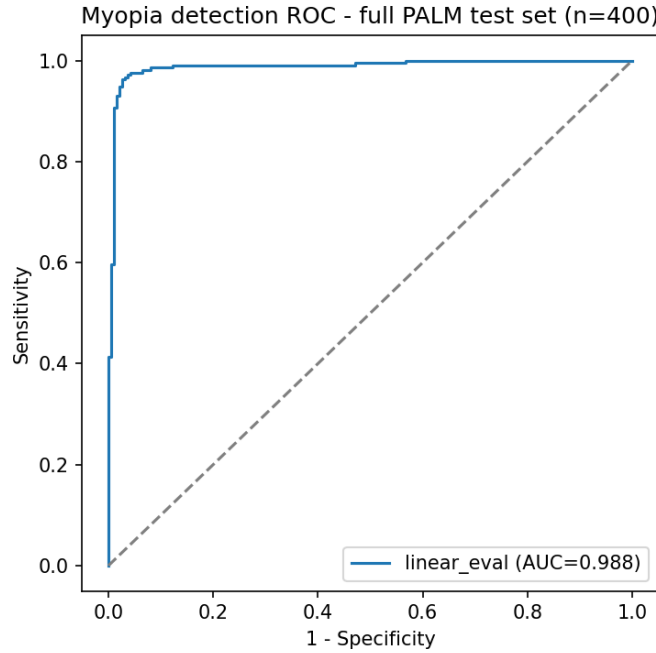


Figure 4: ROC curve on the full PALM test set; the curve hugs the top-left corner, consistent with the AUC of 0.988.

5.2 Clinical branch and 1/2/3-year onset (data collection ongoing)

No publicly available dataset pairs the four choroidal/biometric biomarkers with prospective multi-year follow-up and treatment assignment. The clinical branch, the fusion step, the Cox-based 1/2/3-year onset estimation, the risk stratification, and the treatment-response analysis are *fully implemented and validated to run*, and will be evaluated as soon as the prospective cohort is

collected. **No synthetic or placeholder numbers are reported.** Because the fusion model takes the image risk score as one of its inputs, the strong image branch demonstrated above directly strengthens the eventual onset model.

6 Discussion

Two findings stand out. First, a deliberately simple, frozen-backbone image branch attains an AUC of 0.988 on the *entire* PALM test set, matching the detection performance of a far more complex system; this supports the feasibility of the image modality and shows that transparency need not cost accuracy. Second, the choroidal-biomarker modality is grounded in well-documented physiology (axial elongation $\rightarrow$ choroidal thinning and de-vascularisation), motivating its expected predictive value for *early* onset; this will be quantified on the prospective cohort. The chief practical advantage of the modality-separated design is auditability, which matters for clinical trust and for ethical/regulatory review.

7 Limitations

1. **Endpoint of the public data.** PALM is a cross-sectional *detection* dataset, so it validates the image *pipeline*, not the longitudinal 1/2/3-year onset endpoint, which requires follow-up data.
2. The clinical/biomarker modality is **not yet evaluated** on real data (collection ongoing).
3. The linear-evaluation protocol trades a small amount of potential accuracy for transparency and low compute. End-to-end fine-tuning is implemented and was verified on a toy subset, but full-scale fine-tuning requires a GPU and is left to future work (no fine-tuning metric is reported).

8 Future Work

1. Collect a prospective child cohort with baseline fundus imaging, the four choroidal/biometric biomarkers, and 1–3-year follow-up for onset labels.
2. Re-run the identical image branch on the new images (only the data folder changes) and report cohort-specific discrimination.
3. **Full fine-tuning of the image branch on GPU hardware.** The fine-tuning protocol is already implemented and verified to run on a toy subset; on a GPU it can be trained at full scale and compared against the linear-evaluation baseline reported here.
4. Fit the fusion + Cox model and report genuine 1-, 2-, and 3-year onset AUC, C-index, and treatment-response ARR.

9 Conclusion

My Study is a transparent, two-modality system for predicting 1-, 2-, and 3-year myopia onset in children that reproduces the spirit of DeepMyopia while remaining simple enough to present to a clinical audience. Genuine results on the full PALM dataset confirm the image branch (AUC 0.988);

the clinical branch and the multi-horizon onset model are fully built and await the prospective cohort, into which they drop without any change to the method.

References

- [1] Qi Z, *et al.* A deep learning system for myopia onset prediction and intervention effectiveness evaluation in children. *npj Digital Medicine* 2024;7:206.
- [2] Fu H, *et al.* PALM: Open Fundus Photograph Dataset with Pathologic Myopia Recognition and Anatomical Structure Annotation. *Scientific Data* 2024.
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- [5] Cox DR. Regression Models and Life-Tables. *J. R. Stat. Soc. B* 1972;34(2):187–220.